

Name: Tieu Nghi

TRANSPORT PHENOMENA I, CHE 311

QUIZ 2

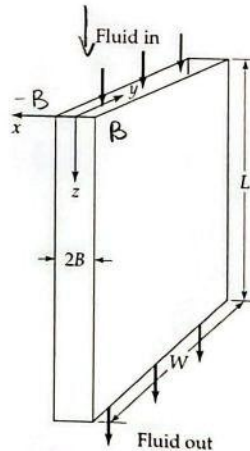
CLOSED BOOK, TIME: 15 MINUTES

A Newtonian fluid is in laminar flow through a narrow slit formed by two vertical parallel walls a distance $2B$ apart. The wall at $x = -B$ is stationary, while the wall at $x = +B$ moves with a constant velocity v_B in z direction.

Make a shell momentum balance to obtain the velocity profile $v_z(x)$.

Model assumption:

- Steady state
- Neglect edge effects, assuming $B \ll W \ll L$
- Constant density ρ and viscosity μ
- $p(z=0) = p_0$, $p(z=L) = p_L$, and $\frac{\partial p}{\partial x} = \frac{\partial p}{\partial y} = 0$
- $v_z = v_z(x)$, and $v_x = v_y = 0$.



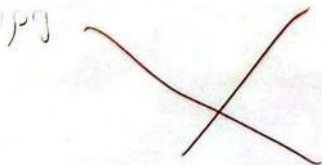
- v_B in z -direction - $\Delta x | \phi_{xz} |_{x=0} = 0$?
- steady state : $v_x = 0 @ x = 0$.

$$\mu \quad v_B = p + \rho g h = p_0 + \rho g L$$

$$\frac{dv_z}{dx} = \frac{d}{dx} (p_0 + \rho g L) = \frac{dp_0}{dx} + \frac{d\rho g L}{dx}$$

$$= \rho g \frac{dL}{dx} = -\mu \rho g L (\Delta x) \Big|_{-B}^B = -\mu \rho g L (-B - B) = 2B \mu \rho g L$$

shell momentum balance obtain the velocity profile $v_z(x) : +2B \mu \rho g L$.



10/100